

UCG Fiber DNS Catch-All Hardening Guide

Device: UCG Fiber | UniFi Network: 10.4.57 | Scope: Default, Trusted, and IoT networks only

Goal: force hardcoded plaintext DNS to your gateway, then have the gateway forward upstream through your encrypted DNS provider. For encrypted client-side DNS, block it rather than trying to redirect it.

1. Current Network Map

Network	Gateway DNS	Catch Rule Needed
Default	192.168.1.1 on br0	DNS Catch – Default
Trusted	192.168.10.1 on br10	DNS Catch – Trusted
IoT	192.168.20.1 on br20	DNS Catch – IoT

Do not add Guest rules if you are not using the Guest network.

2. Confirm Gateway Upstream Encrypted DNS

1. Open UniFi Network.
2. Go to **Settings** and confirm DNS Shield / encrypted DNS is enabled for the gateway.
3. Confirm the selected upstream provider is your intended provider, such as your NextDNS profile.
4. Confirm DHCP on each network advertises the gateway IP as DNS:
 - Default: 192.168.1.1
 - Trusted: 192.168.10.1
 - IoT: 192.168.20.1

3. Verify the Three NAT Catch Rules

Path for Network 10.x: **Settings > Policy Table > Create New Policy > NAT**. If the rule already exists, open it and verify the fields below.

Field	Default	Trusted	IoT
Name	DNS Catch – Default	DNS Catch – Trusted	DNS Catch – IoT
NAT Type	Destination NAT	Destination NAT	Destination NAT
Interface	br0 / Default	br10 / Trusted	br20 / IoT
Protocol	TCP and UDP	TCP and UDP	TCP and UDP
Destination Port	53	53	53
Translated IP	192.168.1.1	192.168.10.1	192.168.20.1
Translated Port	53	53	53
IP Version	IPv4 only	IPv4 only	IPv4 only

Do not create NAT redirects for 853 or 443. DNS-over-TLS, DNS-over-QUIC, and DNS-over-HTTPS use encryption and certificate validation. Redirecting them to port 53 usually breaks traffic unpredictably. Block them instead.

4. Do Not Use Object Manager for Port Blocking

In Network 10.4.57, **Object Oriented Networking / Object Manager** is outcome-based. It is useful for devices, networks, IP addresses, domains, regions, apps, routing, QoS, and broad internet/local security outcomes. It is not where you define destination-port blocking.

If you created an Object Manager item named `DNS Plaintext`, disable or delete it. That object does not represent TCP/UDP port 53. It can create broader secure/blocklist behavior than intended.

Use the NAT rules from Step 3 for plaintext DNS redirection. Use Policy Table firewall/traffic rules for destination ports such as `853`, `784`, `8853`, and selected `443` DoH destinations.

5. Create Only the Address Objects You Need

Create address objects only for destination IP lists, not for port lists. The main useful object here is the public DoH resolver destination list.

Name	Type	Purpose
Public DoH Resolvers – Block	IP Address / Address Group	Destinations to block on TCP/UDP <code>443</code> because they are known public DoH providers.

Create an address object named `Public DoH Resolvers – Block`. Start with these common resolver IPs and update the group over time:

`1.1.1.1`, `1.0.0.1`, `1.1.1.2`, `1.0.0.2`, `1.1.1.3`, `1.0.0.3`, `8.8.8.8`, `8.8.4.4`, `9.9.9.9`,
`149.112.112.112`, `208.67.222.222`, `208.67.220.220`, `94.140.14.14`, `94.140.15.15`,
`185.228.168.9`, `185.228.169.9`, `76.76.2.0`, `76.76.10.0`, `45.90.28.0`, `45.90.30.0`,
`194.242.2.2`, `194.242.2.3`

This list is a starter set, not a permanent complete database. Use UniFi traffic logs, NextDNS logs, and client testing to expand it.

6. Add Firewall Blocks Above General Allow Rules

Go to **Settings > Policy Table**, not Object Manager. Create firewall/security policies from each client network zone to Internet/WAN. Put these above broad allow-to-internet rules.

If your Policy Table editor exposes a **Port**, **Destination Port**, **Service**, or **Advanced** field, enter the ports directly in the rule. If it does not expose a port/service field, use the legacy firewall/traffic rule editor for these port-specific blocks; Object Manager cannot express them.

Rule A: Block DoT and DoQ

Name	Block Encrypted DNS Ports – Default , Trusted , IoT
Action	Block / Drop
Source	Default network, then repeat for Trusted and IoT
Destination	Internet / WAN
Destination Ports	TCP 853 , UDP 853 , UDP 784 , UDP 8853
Logging	Enable for the first few days, then disable if noisy.

Rule B: Block Known DoH Resolvers on HTTPS

Name	Block Public DoH Resolvers – Default , Trusted , IoT
Action	Block / Drop
Source	Default network, then repeat for Trusted and IoT
Destination	Public DoH Resolvers – Block
Destination Ports	TCP/UDP 443
Logging	Enable initially to discover hardcoded devices.

Rule C: Backup Block for Plain DNS to WAN

Your DNAT rules should catch TCP/UDP 53 before this rule. This backup rule confirms that any plain DNS escaping NAT gets blocked.

Name	Block Direct DNS to WAN – Default , Trusted , IoT
Action	Block / Drop
Source	Default network, then repeat for Trusted and IoT
Destination	Internet / WAN
Destination Ports	TCP/UDP 53
Placement	Below the NAT rules, above broad allow-to-internet policy.

7. Optional Hardening

- Block VPN/proxy/Tor application categories for IoT if you do not need them.
- Keep QUIC generally allowed unless you specifically want to force HTTP/3 off. If blocking QUIC, start with UDP 443 only to the DoH resolver IP group.
- If you later enable IPv6 on LAN networks, add equivalent IPv6 DNS policy or disable IPv6 RA/DHCPv6 on those VLANs. IPv4 DNAT does not control IPv6 DNS.

8. Validation Checklist

1. From a Default client, run a lookup against a hardcoded resolver, for example `nslookup example.com 8.8.8.8`. It should still resolve because NAT redirects it to the gateway.
2. Repeat from Trusted and IoT clients.
3. Check UniFi policy counters for the three DNS catch NAT rules. UDP counters should increase; TCP may stay low unless a client uses TCP DNS.
4. Try a DoT client against `1.1.1.1:853` or `8.8.8.8:853`. It should fail.
5. Try a DoH endpoint such as Cloudflare or Google. It should fail when the destination IP is in your DoH block group.
6. Open your encrypted DNS provider dashboard and confirm normal client browsing is visible through the gateway profile.
7. Watch UniFi logs for blocked devices. If a device is repeatedly hitting a missing DoH resolver IP, add that IP to the DoH block group.

9. Rollback Plan

1. Disable the new firewall block rules first; leave the NAT catch rules in place.
2. If a specific device breaks, review its blocked destination and decide whether to add a narrow exception.
3. Only disable NAT catch rules if basic DNS resolution fails across an entire VLAN.

References

- Ubiquiti Help Center: DNAT, SNAT, and Masquerading in UniFi.
- Ubiquiti Help Center: Traffic & Policy Management in UniFi.
- Ubiquiti Help Center: Content and Domain Filtering in UniFi.